Advanced Data Preprocessing Project

Project Title: Employee Performance Data Preparation and Exploratory Data Analysis

Phase 1 - Dataset Generation with NumPy (Steps 1-10)
1. Generate 500 employee records.
2. Create unique EmployeeID.
3. Generate Age.
4. Generate Experience based on Age.
5. Generate Salary using Experience and Department.
6. Generate Training Hours.
7. Generate Projects Completed.
8. Generate Attendance Rate.
9. Generate Department and Education.
10. Compute Performance Score using multiple variables.

Phase 2 - Build DataFrame (11-15)
11. Convert arrays to DataFrame.
12. Rename columns.
13. Reorder columns.
14. Set EmployeeID as index.
15. Display structure.

Phase 3 - Dataset Inspection (16-23)
16. Head(). 17. Tail(). 18. Shape. 19. Data types. 20. Describe(). 21. Value counts. 22. Unique values. 23. Memory usage.

Phase 4 - Data Quality (24-33)
24. Insert missing values.
25. Insert duplicates.
26. Insert inconsistent categories.
27. Insert invalid values.
28. Detect missing values.
29. Handle missing values.
30. Remove duplicates.
31. Standardize categories.
32. Detect invalid values.
33. Correct invalid values.

Phase 5 - Feature Engineering (34-41)
34. AgeGroup.
35. SalaryLevel.
36. ExperienceLevel.
37. TrainingCategory.
38. AttendanceCategory.
39. HighPerformer flag.
40. BonusEligibility.
41. Rearrange columns.

Phase 6 - Data Analysis (42-49)
42. Filter records.
43. Sort records.
44. GroupBy Department.
45. Aggregate statistics.
46. Pivot Table.
47. Crosstab.

48. Correlation Matrix.
49. Interpret strongest relationships.

Phase 7 - Visualization (50-57)
Histogram, Bar Chart, Pie Chart, Scatter Plot, Box Plot, Count Plot, Violin Plot, Correlation Heatmap.

Final Questions
1. Which department has the highest average performance?
2. Which department has the highest salary?
3. Which education level performs best?
4. Which age group performs best?
5. Is attendance related to performance?
6. Do more projects always improve performance?
7. Which variables have the strongest positive correlation?
8. Which variables have the strongest negative correlation?
9. Which visualization provided the most useful insight?
10. Which preprocessing step improved data quality the most?
11. Identify three important patterns.
12. Provide five evidence-based recommendations.